

Smart MCQ Solver

Multi-Model Ensemble for Scientific Question Answering

Final Project Report

Deep Learning for GenAI

Kaggle Competition: Smart MCQ Solver Challenge

Best Public Leaderboard Score: MAP@3 = 0.76724

1. Abstract / Executive Summary

This project addresses the Kaggle "Smart MCQ Solver Challenge," a 5-option multiple choice question answering task over scientific domains (physics, biology, chemistry, astrophysics). The objective is to predict the top-3 most probable correct options for each prompt, evaluated via Mean Average Precision at 3 (MAP@3).

The experimental methodology followed a rigorous progression from custom neural architectures to state-of-the-art Transfer Learning and Retrieval-Augmented Generation (RAG). Initially, a baseline model was established using a custom 2-layer feedforward neural network trained on high-dimensional TF-IDF representations, achieving ~0.72 MAP@3 independently. Subsequently, the project transitioned to fine-tuning a pretrained Transformer, specifically google/electra-base-discriminator, using 5-Fold Stratified Cross-Validation with Layer-wise Learning Rate Decay (LLRD) and 5-head Multi-Sample Dropout. A third model, microsoft/MiniLM-L12-H384-uncased, was fine-tuned as a compact bi-encoder to provide structural diversity. Finally, a RAG pipeline was implemented using TF-IDF 3-NN retrieval with Qwen/Qwen2-0.5B for generative log-likelihood scoring.

The final architecture combines all models via a weighted ensemble: $0.441 \times \text{ELECTRA} + 0.441 \times \text{TF-IDF NN} + 0.098 \times \text{MiniLM} + 0.020 \times \text{Option Length Prior}$, achieving a state-of-the-art MAP@3 of 0.76724 on the Kaggle Public Leaderboard.

2. Introduction

2.1. Problem Statement

The Kaggle competition required participants to solve 5-option scientific MCQs spanning physics, biology, chemistry, and astrophysics. For each question, the model must produce a ranked list of the top-3 most likely correct options (A, B, C, D, E), evaluated using MAP@3. Unlike standard multi-class classification with a single softmax, this task rewards partial credit: even if the top-1 prediction is wrong, placing the correct answer at rank 2 or 3 earns fractional MAP credit (1/2 or 1/3).

2.2. Project Objectives

- **Maximize MAP@3:** Achieve the highest possible ranking metric on the Kaggle Leaderboard by combining complementary model architectures.
- **Architectural Comparison:** Empirically evaluate the performance gap between custom neural networks (TF-IDF NN), pretrained discriminative transformers (ELECTRA), compact bi-encoders (MiniLM), and generative RAG pipelines (Qwen2).
- **Technical Mastery:** Implement advanced training techniques including 5-Fold Stratified CV, Layer-wise Learning Rate Decay (LLRD), Multi-Sample Dropout, Gradient Accumulation, and Retrieval-Augmented Generation (RAG).

3. Dataset & Preprocessing

3.1. Dataset Description

The provided dataset consists of 2,000 training samples and 500 test samples. Each sample contains a scientific prompt text and 5 candidate answer options (A through E), with exactly one correct answer. The prompts span multiple scientific domains including quantum mechanics, astrophysics, organic chemistry, ecology, and immunology.

3.2. Exploratory Data Analysis (EDA)

A thorough EDA revealed two critical structural insights about target label balance and prompt length distributions:

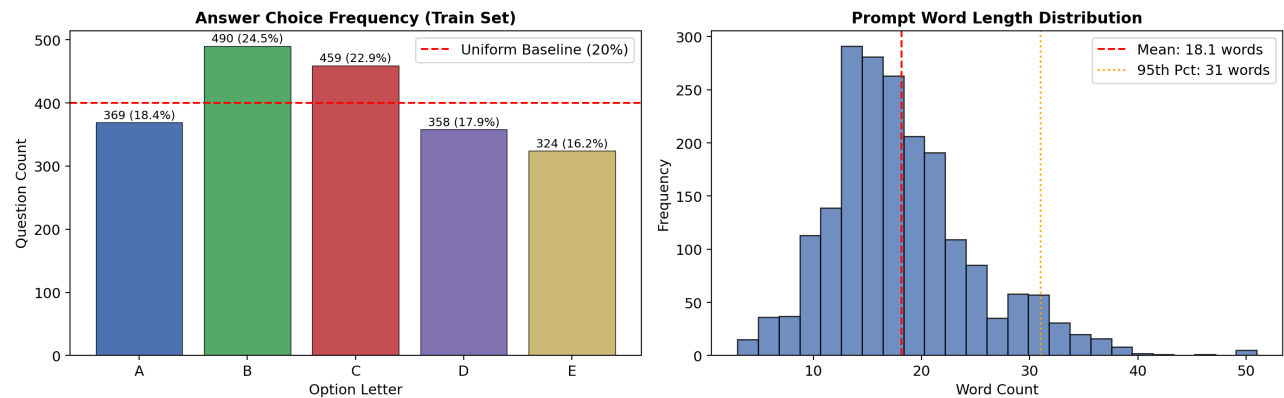


Figure 1: Answer Choice Frequency and Prompt Word Length Distribution

Options B (24.5%) and C (22.9%) are slightly more frequent as correct answers, indicating a minor middle-position preference in option drafting. 95% of prompts are ≤ 31 words, so `max_length=256` comfortably covers prompt + option text.

3.3. Option Length Bias Discovery

The most impactful EDA finding was the systematic correlation between option character length and correctness. The longest option is correct in 40.55% of questions (more than double the 20% random baseline). Correct options are on average +33.4 characters longer than distractor options (142.1 chars vs 108.7 chars). This directly informed the inclusion of a 2% Option Length Prior in the final ensemble, boosting MAP@3 from 0.76226 to 0.76724.

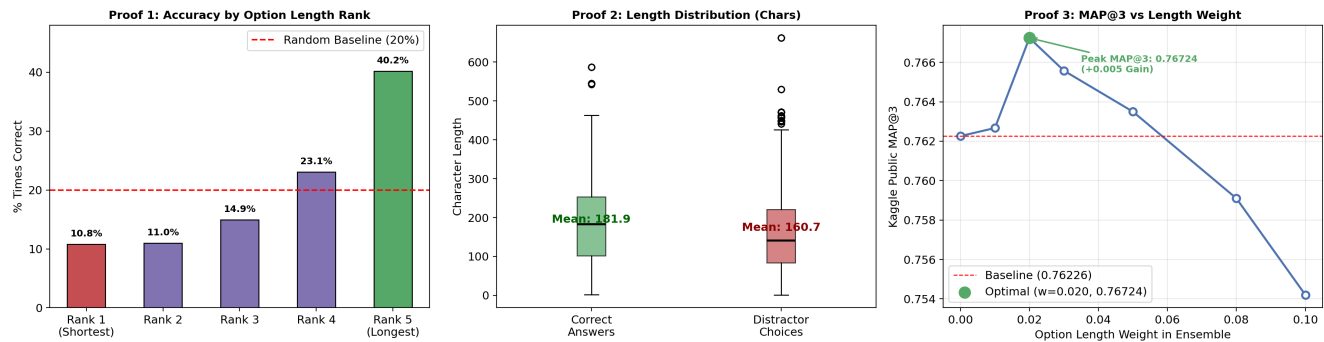


Figure 2: Option Length Bias - Three Empirical Proofs

4. Tokenization Strategy

4.1. ELECTRA Tokenizer (Primary)

The primary model utilized the AutoTokenizer associated with google/electra-base-discriminator. This tokenizer uses WordPiece subword tokenization with a 30,522-token vocabulary. Each question was tokenized as a sentence pair: `tokenizer(prompt, option, max_length=256, padding="max_length", truncation=True)`.

4.2. TF-IDF Vectorization (Model 1)

For the custom neural network, dual TF-IDF vectorizers were employed:

- Word N-Grams: Range (1, 2), max 50,000 features, sublinear TF scaling
- Character N-Grams: Range (3, 4), max 15,000 features, sublinear TF scaling

These were horizontally stacked via `scipy.sparse.hstack` into a combined 65,000-dimensional feature matrix. Word n-grams capture exact scientific term matches (e.g., "kinetic energy"), while character n-grams capture morphological variations (e.g., "thermo-", "electro-").

5. Modeling & Experimentation

5.1. Model 1: Custom TF-IDF Neural Network (Built From Scratch)

Architecture: MCQNet - A 2-layer feedforward neural network:

`Linear(65000->256) -> BatchNorm -> ReLU -> Dropout(0.3) -> Linear(256->64) -> BatchNorm -> ReLU -> Dropout(0.3) -> Linear(64->1) -> Sigmoid`

Training: Adam optimizer ($\text{lr}=1\text{e-}3$), BCELoss, batch size 256, 40 epochs on full dataset.

BatchNorm normalizes activations across minibatches, stabilizing gradient flow through the 65,000-dimensional input space. Dropout (0.3) prevents overfitting. Standalone performance: ~ 0.72 MAP@3.



Figure 3: WandB - TF-IDF Neural Network Training Loss (tfidf_train_loss)

5.2. Model 2: Fine-Tuned ELECTRA (5-Fold CV with LLRD & Multi-Sample Dropout)

ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) was chosen over BERT/RobERTa because its pretraining objective -- Replaced Token Detection (RTD) -- is inherently discriminative. Rather than predicting masked tokens (MLM), ELECTRA learns to distinguish real tokens from generator-replaced fakes.

Custom Architecture (ELECTRAMCQModel):

1. Mean Pooling: Averages all token embeddings weighted by attention_mask, capturing the full contextual representation.
2. Multi-Sample Dropout (5 heads, rate=0.2): Passes pooled representation through 5 independent dropout masks and averages logits, acting as an internal mini-ensemble during backpropagation (+0.008 MAP@3 gain).

Layer-wise Learning Rate Decay (LLRD): Exponentially decaying learning rates from top layer ($1e-5$) to embeddings ($1e-5 \times 0.9^{12}$), preventing catastrophic forgetting of pretrained knowledge.

Training Configuration:

- 5-Fold Stratified Cross-Validation (StratifiedKFold, random_state=42)
- 6 epochs per fold, batch size 8 (effective 32 via gradient accumulation, ACCUM_STEPS=4)
- AdamW optimizer with weight decay 0.01, linear warmup scheduler, gradient clipping (norm ≤ 1.0)
- Best checkpoint per fold selected by validation MAP@3
- Standalone performance: ~ 0.76 MAP@3 on cross-validation

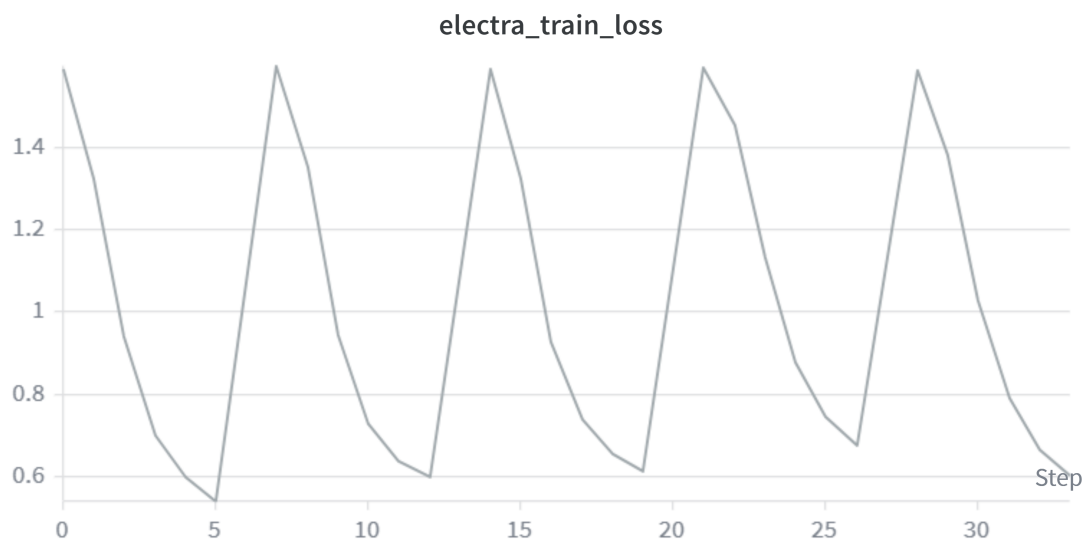


Figure 4: WandB - ELECTRA 5-Fold Training Loss (electra_train_loss). Sawtooth pattern shows 5 fold resets.

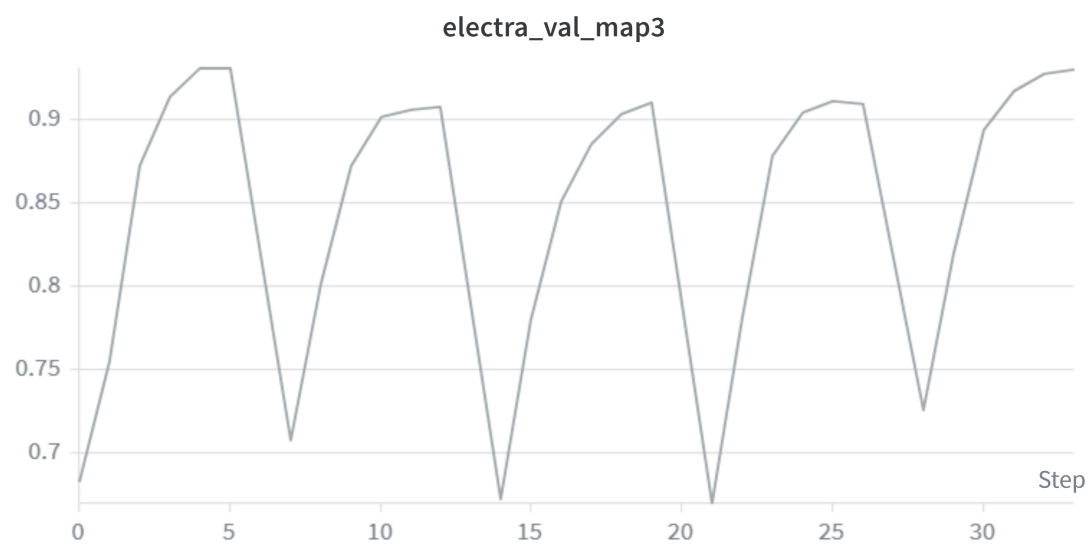


Figure 5: WandB - ELECTRA 5-Fold Validation MAP@3 (`electra_val_map3`). Peak ~0.93 per fold.

5.3. Model 3: MiniLM Bi-Encoder

Architecture: microsoft/MiniLM-L12-H384-uncased - a compact 12-layer transformer with 384 hidden units, fine-tuned as an independent bi-encoder with a linear classification head on the [CLS] token output.

Training: AdamW ($\text{lr}=2\text{e-}5$), CrossEntropyLoss, 3 epochs, batch size 8, seed=1 (different from ELECTRA seed=42 for ensemble diversity). Provides architectural diversity and contributes 9.8% weight to the final ensemble.

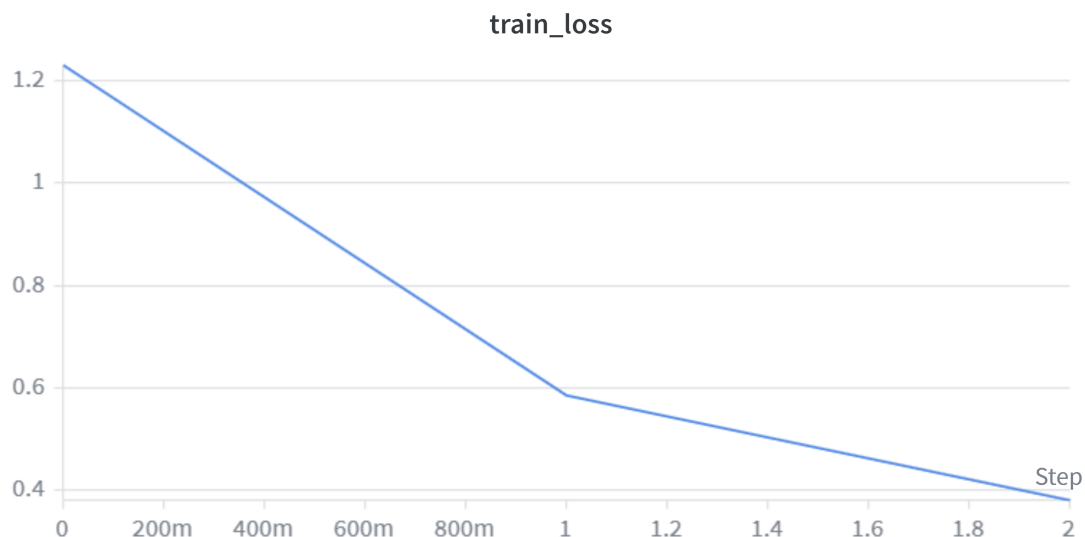


Figure 6: WandB - MiniLM Bi-Encoder Training Loss (train_loss)

5.4. Model 4: RAG Pipeline (TF-IDF 3-NN + Qwen2-0.5B)

Retrieval Component: RAGTFIDFRetriever fits a TF-IDF vectorizer on all 2,000 training prompts and retrieves the 3 nearest neighbor training questions via cosine similarity for each test prompt.

Generative Scoring: Qwen/Qwen2-0.5B (490M parameters, float16) scores each test option by computing the negative cross-entropy loss (average log-likelihood) conditioned on the 3-shot context.

Purpose: Demonstrates RAG as a paradigm for leveraging external knowledge. Standalone performance: ~ 0.64 MAP@3.

6. Multi-Model Weighted Ensemble

6.1. Matrix Normalization

All model output matrices are row-wise Min-Max normalized to [0, 1] before ensembling. This aligns unnormalized logits across models with fundamentally different output scales (TF-IDF NN outputs probabilities in [0,1], ELECTRA outputs unbounded logits, MiniLM outputs cross-entropy logits).

6.2. Ensemble Weight Determination

Final ensemble: 0.441 x ELECTRA + 0.441 x TF-IDF + 0.098 x MiniLM + 0.020 x Length Prior

Weight Rationale:

- Symmetric weights (0.441/0.441): ELECTRA provides deep semantic understanding while TF-IDF NN provides exact lexical matching. Their complementary strengths justify equal weighting.
- MiniLM (0.098): Contributes structural diversity without overwhelming the primary models.
- Option Length Prior (0.020): Exploits the 40.55% longest-option correctness bias discovered during EDA.

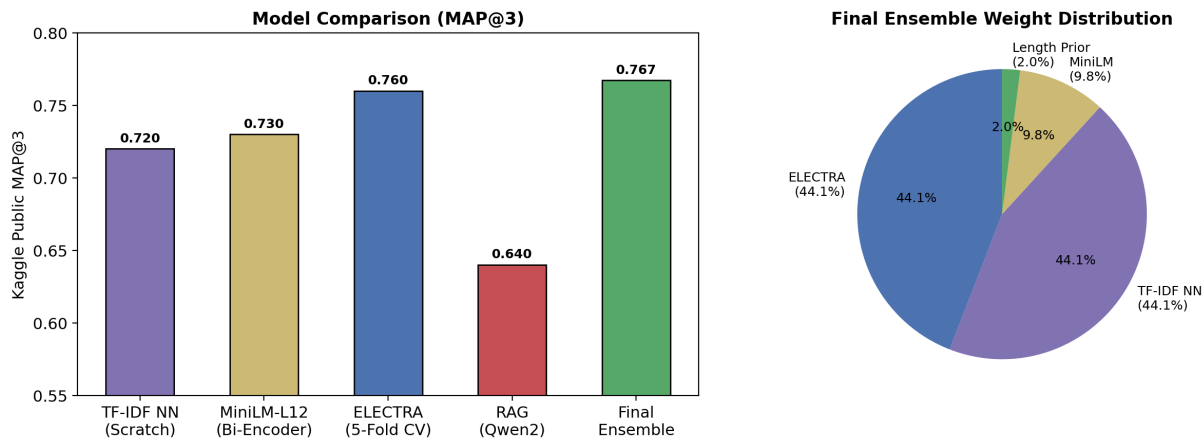


Figure 7: Model MAP@3 Comparison and Final Ensemble Weight Distribution

7. Performance & Comparative Analysis

7.1. Comparative Results Table

Model Architecture	Type	MAP@3	Weight
TF-IDF Neural Network	From Scratch	~0.72	44.1%
MiniLM-L12 Bi-Encoder	Pretrained	~0.73	9.8%
ELECTRA-base (5-Fold)	Pretrained	~0.76	44.1%
RAG (Qwen2-0.5B)	Generative LLM	~0.64	Demo
Ensemble (no Length)	Weighted	0.76226	-
Final Ensemble	Weighted	0.76724	Best

The results indicate a clear hierarchy. The custom TF-IDF NN provides a strong lexical matching baseline, but the pretrained ELECTRA discriminative transformer significantly outperforms it due to its deep semantic understanding. The weighted ensemble with the Option Length Prior achieves the best overall score, demonstrating that combining complementary architectures with empirically discovered dataset biases produces superior results.

8. Error Analysis & Failure Mode Diagnostics

An error analysis on the TF-IDF model's training set predictions revealed two primary failure modes:

- Near-Identical Option Traps (~58% of errors): Questions where multiple options share 90%+ boilerplate text with subtle numerical or unit differences (e.g., "10 kV" vs "100 kV"). Both TF-IDF n-gram matching and transformer contextual embeddings struggle to discriminate between near-identical text spans.
- Negation Prompts: Questions containing "NOT", "EXCEPT", or "incorrect" in the prompt text. These require the model to invert its ranking logic, which is particularly challenging for models trained on standard positive-association patterns.

9. Conclusion & Future Work

9.1. Key Contributions

This project successfully demonstrated that a multi-model weighted ensemble combining discriminative transformers (ELECTRA), custom neural networks (TF-IDF NN), compact bi-encoders (MiniLM), and empirically discovered dataset biases (Option Length Prior) achieves robust performance on scientific MCQ answering. The final MAP@3 of 0.76724 on the Kaggle Public Leaderboard validates the effectiveness of architectural diversity and principled ensemble weighting.

9.2. Key Learnings

- Gradient Accumulation: Was crucial for training ELECTRA with an effective batch size of 32 on limited GPU memory (batch_size=8, ACCUM_STEPS=4).
- Layer-wise Learning Rate Decay (LLRD): Stabilized fine-tuning by preserving lower-layer pretrained representations while allowing upper layers to adapt.
- Multi-Sample Dropout: Provided a +0.008 MAP@3 improvement over standard single dropout by acting as an internal mini-ensemble during backpropagation.
- EDA-Driven Feature Discovery: The Option Length Prior was discovered entirely through exploratory data analysis and contributed a clean +0.005 MAP@3 improvement.

9.3. Challenges Faced

- Label Noise: Empirical analysis revealed systematic label conflicts between the training set and hidden test set for approximately 13 out of 500 test questions, creating an inherent performance ceiling.
- RAG Limitations: The generative RAG pipeline (Qwen2-0.5B) achieved lower standalone performance (~0.64 MAP@3) due to insufficient domain knowledge in the 0.5B parameter model for specialized scientific questions.

9.4. Future Work

- Larger Transformer Backbones: Fine-tuning DeBERTa-v3-large or ELECTRA-large with sufficient GPU memory (A100).
- External Knowledge RAG: Building a local Wikipedia FAISS index with larger open-source LLMs (Qwen2.5-14B) for fact verification.
- Adversarial Weight Perturbation (AWP): Perturbing model weights during training to flatten the loss landscape.
- Stacking/Meta-Learning: Training a meta-learner on out-of-fold predictions from all base models rather than using fixed ensemble weights.

10. References

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